

Focal Point

Infrastructure: Power Grid

Budget: ₹50

Location: Ludhiana, INdia

Target Year: 2040

Risk Level: N/A

Risk Score: 76/100

Climate Resilience and Engineering Report for Ludhiana Power Grid

Introduction

The city of Ludhiana, located in the state of Punjab, India, is prone to flooding due to its geographical location. The power grid infrastructure within the specified coordinates [30.8898, 75.6555], [30.8898, 75.6596], [30.8869, 75.6593], [30.8875, 75.6558] is at risk of damage from flood events. This report provides three site-specific resilience strategies to enhance the flood resilience of the power grid infrastructure in Ludhiana by 2040.

STRATEGY 1

Elevating Substations and Switching Stations

Elevating substations and switching stations above the projected flood levels can prevent damage to critical equipment and minimize power outages. This can be achieved by constructing elevated platforms or relocating the equipment to higher ground. By doing so, the power grid can continue to operate during flood events, ensuring a reliable supply of electricity to the community.

STRATEGY 2

Implementing Flood-Resistant Construction Materials

Using flood-resistant construction materials for power grid infrastructure, such as waterproof concrete and stainless steel, can enhance its durability and resistance to flood damage. These materials can withstand the corrosive effects of floodwaters, reducing the need for repairs and replacements. This strategy can help maintain the structural integrity of the power grid during flood events.

STRATEGY 3

Installing Smart Flood Sensors and Monitoring Systems

Installing smart flood sensors and monitoring systems can provide real-time data on flood levels, enabling proactive measures to be taken to protect the power grid. These systems can detect rising flood levels, triggering alerts and automatic shutdowns of affected sections of the grid, thereby preventing damage and minimizing power outages. This strategy can enhance the responsiveness and resilience of the power grid during flood events.

RESILIENCE SCORE: 85%

